

Lecture 2B - Causal Identification

A Causal Graph Perspective

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① Causal identification

② Summary

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Defining causal identification

Four identification strategies

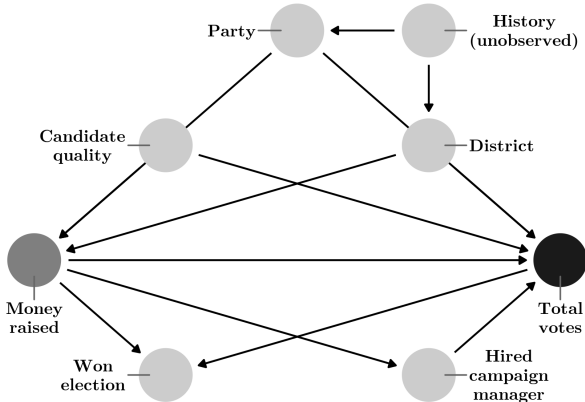
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Defining causal identification

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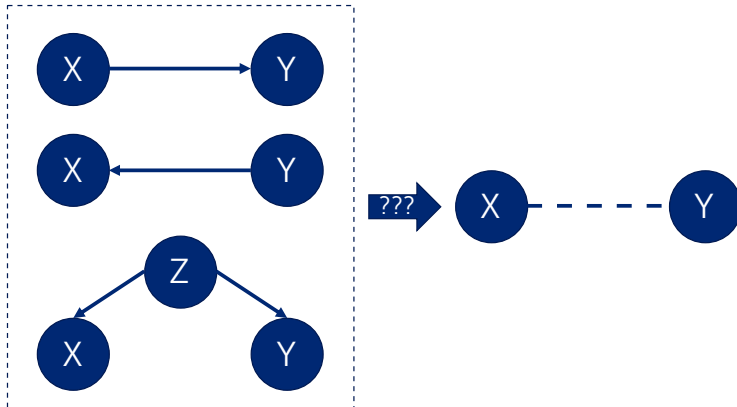
Two types of causal inference

- Type 1: Given data of a system, to find its causal structure.
- **Type 2: Given a causal structure, measure the effect of one variable on another.**



Explaining “Correlation $\neq$ Causation”

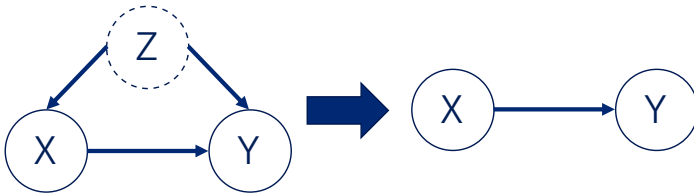
- Correlation is produced by causation
- Multiple causal structures (DAGs) can produce same correlation pattern



The core problem of causal inference

Definition (The Core Problem)

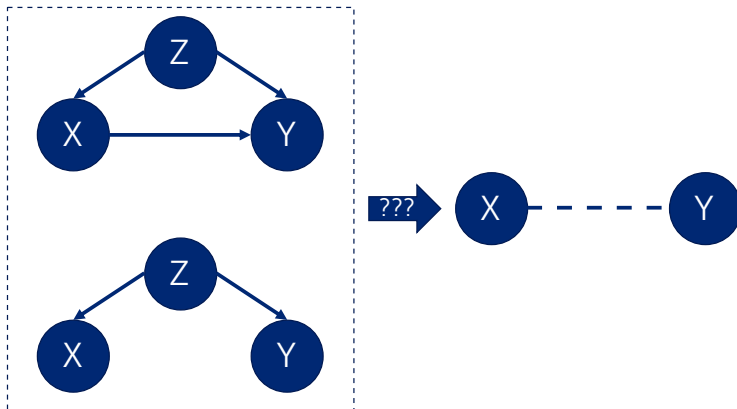
Suppose we want to quantify the effect of X on Y from the DAG below and we perfectly observe X and Y . However, the set of confounders Z is partially or unobserved.



Z has many names: lurking variables, omitted variables, unobservable etc.

The core problem is an identification problem

In almost all business research, obtaining $P(Y \mid \text{do}(X))$ boils down to distinguishing two observation-equivalent DAGs.



Defining the identification problem

Definition (**The identification problem**)

Multiple data-generating processes can result in the same patterns in data.

- **Data-generating processes:** models or theories.
- **Patterns in data:** information in data or statistics of data (e.g., correlations).

Defining the identification problem

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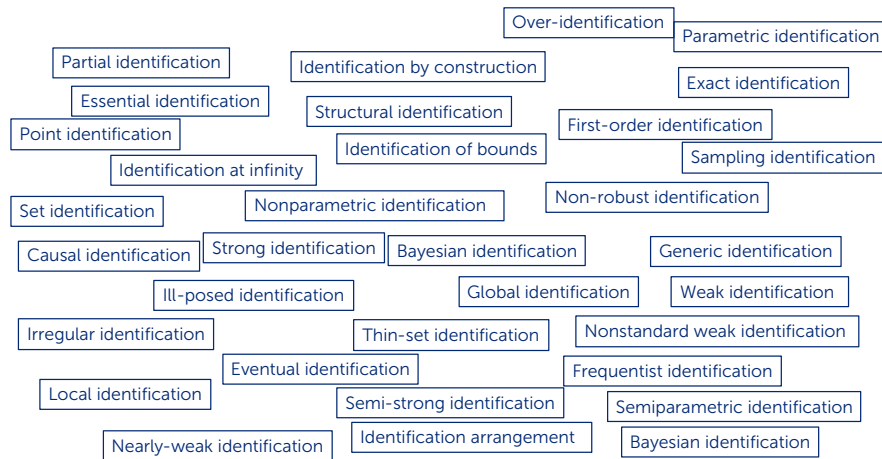
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Endogeneity issue in linear regression $Y = \beta X + \varepsilon$

$$\underbrace{P(Y | X)}_{\text{Patterns in Data}} = P(\beta X + \varepsilon | X) = \underbrace{P(\beta X | X)}_{\text{Cause 1}} + \underbrace{P(\varepsilon | X)}_{\text{Cause 2}}$$

The complex concept of identification

We stick to the core problem of **causal identification** for clarity.



Causal identification as an axiomatization process

What is an axiomatization process?

Example (Who's Chinese? Left or right?)



- Assumption 1: Chinese with two eyes and Japanese with two ears.
 - **Conclusion 1: Left? Right?**

Causal identification as an axiomatization process

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- Assumption 2: Chinese with blue ties and Japanese with red ties.
 - **Conclusion 2: the man on the right!**

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Causal identification as an axiomatization process

What is an axiomatization process?

Example (Who's Chinese? Left or right?)



- Assumption 2: Chinese with blue ties and Japanese with red ties.
 - **Conclusion 2: the man on the right!**
- Assumption 3: Chinese with slicked-backs and Japanese with side-parts.
 - **Conclusion 3: the man on the left!**

Causal identification as an axiomatization process

Definition (Causal identification)

Making assumptions that are sufficient and learning causal relationships from data based on the assumptions.

- Causal identification requires sufficient assumptions.
- Given the same data, different assumptions may lead to different conclusions.
- Alternative sufficient assumptions exist.

Causal identification as an axiomatization process

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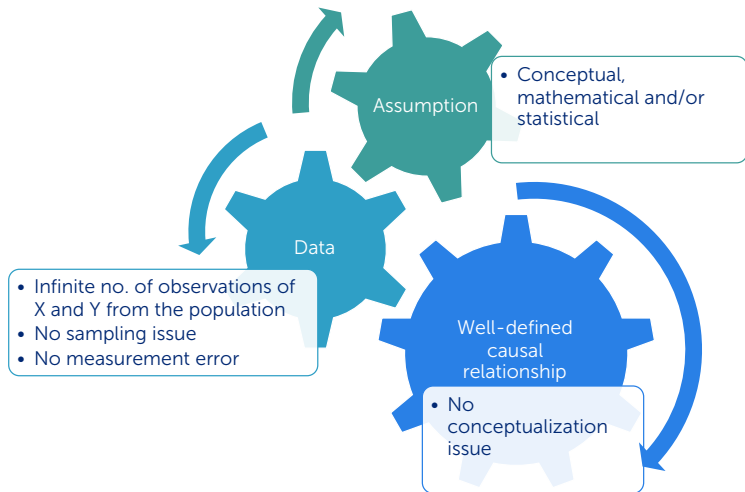
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- Assume $P(\varepsilon | X) = 0$.

3 pillars of causal identification



Identification vs. estimation

Proposition (Identification vs. Estimation)

The relationship between identification and estimation is as such: first, identification logically precedes estimation; and second, one does not logically imply the other.

Two examples

Estimable but not identifiable: $\text{Sales} = \beta \text{Price} + \varepsilon$.

Identifiable but not estimable: Static discrete game with multiple equilibria.

1 Causal identification

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2 Summary

Identification by conditioning

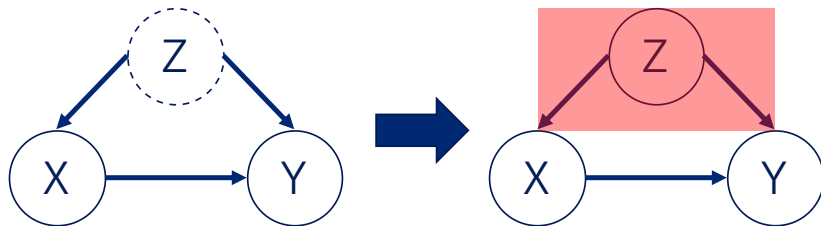


Figure 1: Backdoor criterion

The set of variables Z satisfies the **backdoor assumptions**:

- 1 **Exhaustive**: Z block ALL backdoor paths from X to Y .
- 2 **No colliders**: Z does NOT contain any common descendants (colliders) of X and Y .

Identification by conditioning

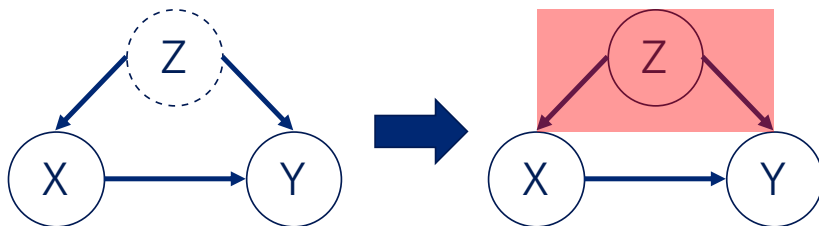


Figure 2: Backdoor Criterion

Implications of the **backdoor assumptions**:

- 1 **Exhaustive**: Z contain all common parents of X and Y.
- 2 **No colliders**: do not include post-treatment variables in Z.

Identification by conditioning

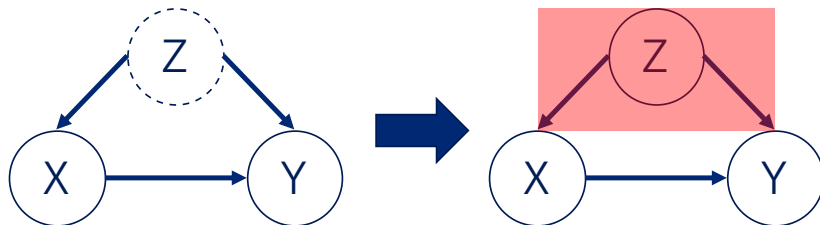


Figure 3: Backdoor Criterion

Estimation of $P(Y \mid \text{do}(X))$:

- Non-parametric estimation:

$$P(Y \mid \text{do}(X)) = \sum_z P(Y \mid X, z) P(z).$$
- In practice, we often use Z as control variables in our regression.

Identification by mechanism

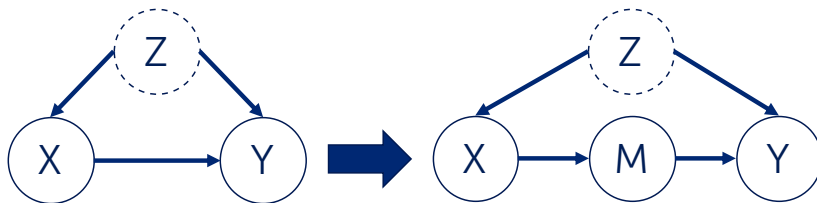


Figure 4: Front-door criterion

The set of variables M must satisfy the **front-door assumptions**:

- **Exogeneity to X :** All backdoor paths of X and M are blocked.
- **Exogeneity to Y :** All backdoor paths of M and Y are blocked by X .
- **Full Mediation:** M intercept all the front-door paths between X and Y .

Identification by mechanism

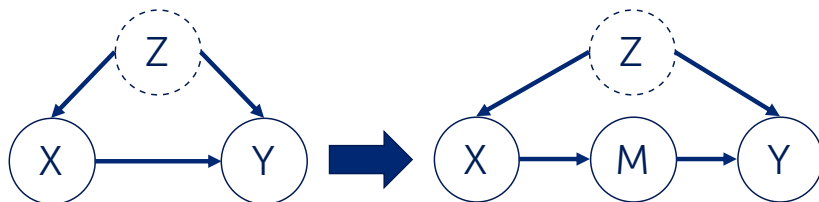


Figure 5: Front-door criterion

Identification of $P(Y \mid \text{do}(X))$ given **front-door assumptions**:

- 1 $P(M \mid \text{do}(X))$ is identified.
- 2 $P(Y \mid \text{do}(M))$ is identified.
- 3 Combining $P(M \mid \text{do}(X))$ and $P(Y \mid \text{do}(M))$.

Identification by mechanism

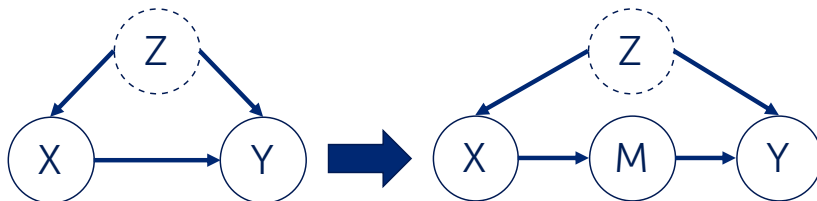


Figure 6: Front-door criterion

Estimation: Combining $P(M \mid \text{do}(X))$ and $P(Y \mid \text{do}(M))$ to get $P(Y \mid \text{do}(X))$.

$$P(Y \mid \text{do}(X)) = \sum_m \left[P(M \mid X) \times \sum_{\tilde{X}} \left(P(Y \mid M, \tilde{X}) P(\tilde{X}) \right) \right]$$

Identification by mechanism

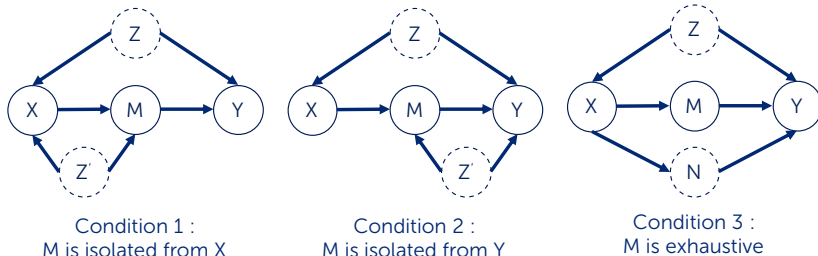
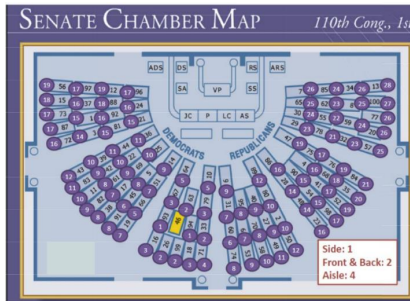


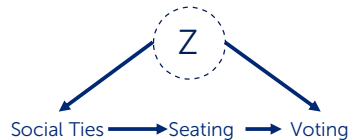
Figure 7: Understanding the front-door assumptions

In practice, you need an **exogenous super-mechanism** as M .

Identification by mechanism: examples



Congressional Seating



The DAG

Table 1: Log-rolling in congressional voting (Cohen and Malloy 2014)

Identification by instruments: the origin

“Exogenous shocks/variation”

- An inventions of economists
- Wide adoption outside of economics
- The origin can be traced back to 1920's

One of the early contributors: Jan Tinbergen (1930)

- Bestimmung und Deutung von Angebotskurven Ein Beispiel
- Determination and Interpretation of Supply Curves: An Example

Identification by instruments: how?

Instead of mutilating the DAG, we expand it!

- $P(Y \mid \text{do}(V)) = P(Y \mid V)$: identified.
- $P(X \mid \text{do}(V)) = P(X \mid V)$: identified.
- To have $P(Y \mid \text{do}(X))$, we solve this equation:
$$P(Y \mid \text{do}(V)) = \sum_X P(Y \mid \text{do}(X)) P(X \mid \text{do}(V)).$$

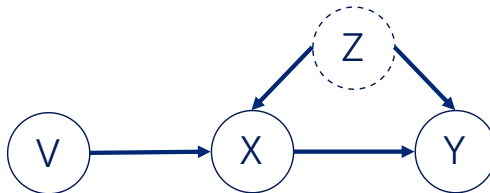
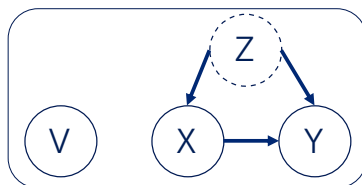
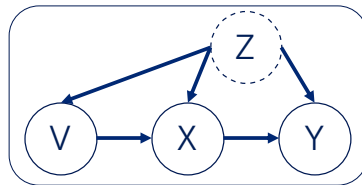


Figure 8: DAG of the IV approach

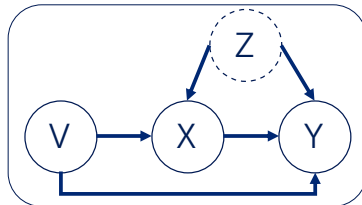
Identification by instruments: assumptions



Relevance



Exogeneity



Exclusion

Is any assumption testable on its own right?

Identification by set

Instead of focusing on a specific value, we try to have a “set”.

- “Half-baked” as compared to point identification.
- a.k.a. partial identification, identification by bounds, or interval identification etc.
- Conceptually insightful but empirically less attractive.

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Example

An experiment with liking of a brand as outcome

Measured with 0 – 9 points scale (0–dislike and 9–like)

Treatment effects must $[-9, +9]$ (“no data bound”)

The law of credibility

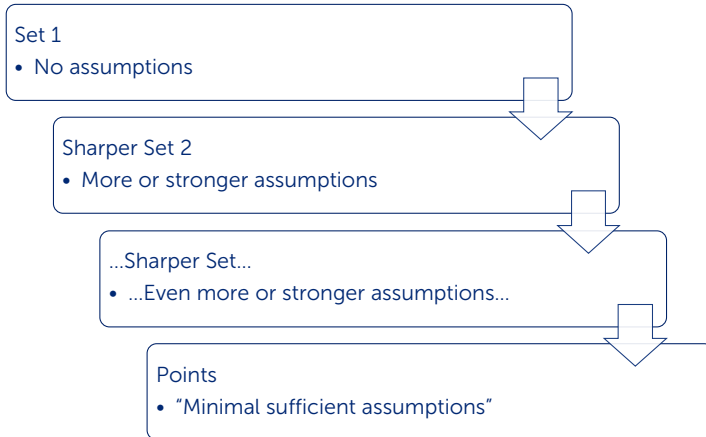
Quote Manski 2003(Chapter 1):

“Empirical researchers should be concerned with both the logic and the credibility of their inferences. Credibility is a subjective matter, yet I take there to be wide agreement on a principle I shall call:

The Law of Decreasing Credibility: The credibility of inference decreases with the strength of the assumptions maintained.

This principle implies that empirical researchers face a dilemma as they decide what assumptions to maintain: Stronger assumptions yield inferences that may be more powerful but less credible.”

Application of the law of credibility



① Causal identification

② Summary

Summary I



The essence of causal inference is identification.



The data alone cannot reveal the underlying causal structures.



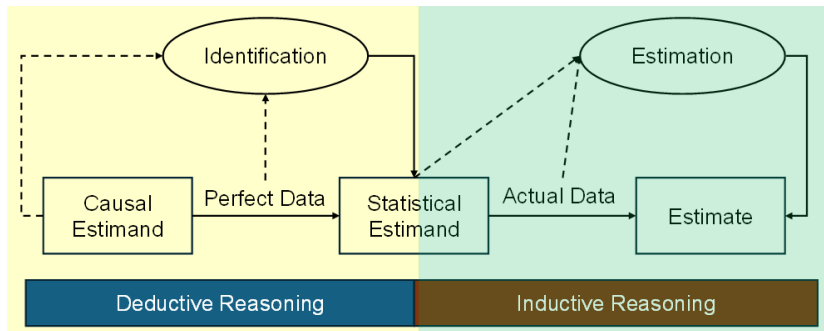
Identification is to find sufficient assumptions to pinpoint a causal structure.



Assumptions + Data = Conclusions

Summary II

The identification-estimation flowchart



References I

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